

Is Quantum Subspace Expansion practical?

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Abstract

Probably not.

Keywords

quantum subspace expansion; variational quantum eigensolver; excited states; active spaces; measurement noise

Abbreviations

QSE, quantum subspace expansion; VQE, variational quantum eigensolver; UCC, unitary coupled cluster; CASCI, complete active space configuration interaction.

Introduction

Computational chemistry has become inseparable from modern chemical research, accelerating the pace at which new compounds, materials, and chemical hypotheses can be evaluated. Electronic-structure calculations underpin much of this progress through molecular-property prediction, high-throughput screening, and atomistic simulation [1–3]. However, systematically accurate wavefunction-based methods become prohibitively expensive for large systems because the dimensions of the exact configuration space grows combinatorially. Practical classical methods like Density Functional Theory (DFT) rely on approximations whose accuracy is not uniformly reliable across different correlation regimes [3]. Quantum computers instead offer an alternative route to modelling nature - which is inherently quantum mechanical - by representing and manipulating quantum states using quantum hardware. Qubits encode and manipulate superpositions of many basis states that would generally require exponentially many amplitudes to represent on a classical machine [4, 5].

The current generation of quantum hardware, albeit rapidly advancing, remains constrained by imperfect gate operations and limited coherent circuit depth. These limitations have motivated hybrid quantum-classical algorithms that use quantum processors for state preparation and measurement, while delegating parameter optimisation and post-processing to classical hardware [6]. Among the most prominent is the Variational Quantum Eigensolver (VQE), which minimises the expectation value of the Hamiltonian to approximate the lowest-energy state accessible to a chosen ansatz [7]. Quantum Subspace Methods (QSM) build on this idea by projecting the full eigenvalue problem onto a compact trial subspace. If the chosen subspace accurately captures the low-energy manifold, the target ground and excited state solutions can be recovered without diagonalising the full many-body Hamiltonian [8].

Quantum Subspace Expansion (QSE), introduced by McClean et al. [9], is a prominent operator-generated realisation of this framework. QSE constructs a trial subspace by applying a selected set of

operators $\hat{O}_\mu$ to a reference state $|\Psi_{\text{ref}}\rangle$, often the VQE ground state. The full many-body Hamiltonian is represented within this subspace through the projected Hamiltonian and overlap matrices,

$$\begin{aligned} H_{\mu\nu} &= \langle \Psi_{\text{ref}} | \hat{O}_\mu^\dagger \hat{H} \hat{O}_\nu | \Psi_{\text{ref}} \rangle, \\ S_{\mu\nu} &= \langle \Psi_{\text{ref}} | \hat{O}_\mu^\dagger \hat{O}_\nu | \Psi_{\text{ref}} \rangle. \end{aligned} \quad (1)$$

The k -th approximate state is then written as

$$|\Psi_k\rangle = \sum_{\mu} c_{\mu k} \hat{O}_\mu | \Psi_{\text{ref}} \rangle. \quad (2)$$

where $\mathbf{c}_k$ is the corresponding vector of expansion coefficients. Applying the Rayleigh-Ritz variational procedure within this trial subspace yields the coefficients and Ritz energies through the generalised eigenvalue problem.

$$\mathbf{H}\mathbf{c}_k = E_k \mathbf{S}\mathbf{c}_k. \quad (3)$$

In general, the operator-generated basis states are not orthogonal, such that $\mathbf{S} \neq \mathbf{I}$. Solving this generalised eigenvalue problem typically requires a whitening transformation containing factors proportional to $\frac{1}{\sqrt{\lambda_i}}$, whereby λ_i are the eigenvalues of $\mathbf{S}$. If the generated basis states are nearly linearly-dependent, $\mathbf{S}$ develops very small or even near-null eigenvalues. Consequently, measurement errors in the matrix elements are strongly amplified, rendering the projected problem numerically unstable. This ill-conditioning of $\mathbf{S}$ is a well-known limitation of QSE, and particularly consequential because statistical shot noises are inherent to real quantum hardware measurements.

Alternative QSM formulations like Quantum Self-Consistent Equation-Of-Motion (Q-SC-EOM) methods avoid this source of instability by constructing an orthogonal excitation manifold [10]. This reduces the overlap matrix to the identity, and consequently the generalised eigenvalue problem into an ordinary one. Doing so has been shown to outperform QSE in spectral accuracy especially when the condition number of $\mathbf{S}$ is large [11]. QSE nevertheless remains of practical interest due to its direct and comparatively flexible formulation. While q-sc-EOM is formulated primarily to recover excitation energy differences relative to a reference state, QSE yields Ritz approximations to absolute energies. QSE therefore serves a dual purpose within the same projected calculation: refinement of the approximate ground state through the lowest Ritz solution, and recovery of excited-state approximations through the higher solutions. [9, 12] Furthermore, QSE does not require its expansion operators to satisfy the vacuum-annihilation condition imposed in self-consistent EOM formulations, allowing a wider range of physically motivated operator pools to be considered. This flexibility leaves substantial scope for tailoring the generated subspace and improving the balance between spectral representability and numerical stability.

Much of the existing QSE literature has evaluated the method using selected molecular systems, often at distorted geometries chosen to probe strongly correlated or numerically challenging regimes. Although these studies have established important limitations in subspace representability and numerical stability, it remains unclear how frequently such limitations arise across chemically diverse molecules under more routine equilibrium conditions. Recent work has also shown that overlap-matrix thresholding can suppress instabilities in the generalized eigenvalue problem. However, the appropriate regularization generally depends on both the underlying system and the level of measurement noise, while overly aggressive thresholding may remove physically relevant subspace directions. A broader characterization is therefore needed to determine how QSE accuracy, representability, and numerical recoverability behave across molecular systems, and how regularization should be applied in practice.

In this work, we address these questions through a systematic benchmark of spin-resolved QSE across 28 chemically varied molecules [13]. Statevector simulations are first used to isolate the intrinsic representability and accuracy of the operator-generated subspace. We compare several reference ansatz classes, ranging from restricted to more expressive constructions, to determine how the prepared reference influences the recoverable spectrum. We then analyse the origins and practical consequences of overlap-matrix ill-conditioning under finite-shot sampling, including its dependence on the overlap spectrum and regularization threshold. Finally, we quantify the associated scaling and measurement requirements, provide practical guidance for threshold selection, and identify directions for improving the construction and numerical treatment of QSE subspaces. The principal contribution of this work is therefore a systematic assessment of when molecular QSE remains both sufficiently expressive and numerically recoverable for practical excited-state calculations.

Results and Discussion

Accuracy and Representability of the QSE Subspace

In this section, we study the spectral accuracy and representability of low-lying roots generated with QSE using statevector simulations. The QSE excitation energies are compared against exact CASCI energies within the same active space. Unless otherwise stated, we focus on the singlet ground state, S0, together with the lowest singlet and triplet excited states, S1, T1 and T2, to characterise QSE performance in the low-energy region of the spectrum.

General Trends in QSE Spectral Accuracy

Table 1: UCC ansatz complexity and median absolute QSE energy errors for the 6e6o active space across the molecular benchmark. Errors are reported in mHa for the lowest singlet ($s0$, $s1$) and triplet ($t1$, $t2$) states.

Ansatz	Ansatz complexity		Median absolute energy error (mHa)			
	Parameters	Circuit depth	$s0$	$s1$	$t1$	$t2$
1UpCCGSDSinglet	30	1306	17.39	35.93	6.70	23.32
2UpCCGSDSinglet	60	2470	6.71	31.39	2.15	5.96
3UpCCGSDSinglet	90	3634	4.14	28.23	2.65	4.30
UCCSDSinglet	54	14 692	0.20	23.69	2.55	3.34
UCCSD	117	12 249	0.14	23.69	2.79	3.17
UCCGSD	315	37 622	0.08	22.25	1.83	2.41

A clear dependence on the choice of reference ansatz is observed, with more expressive ansatz generally producing more accurate QSE spectra. The paired k-UpCCGSDSinglet ansatz restricts excitations heavily to only singles and paired doubles. Consequently, this ansatz exhibits the largest errors in the predicted spectrum. While increasing the number of layers from $k=1$ to $k=3$ systematically improves their performance, the ground-state error at $k=3$ remains more than an order of magnitude larger than that of the next best ansatz. This indicates that increasing the flexibility of the paired ansatz does not fully compensate for the restricts of its excitation manifold. In comparison, UCCSDSinglet and UCCSD achieve similar accuracies across the low-lying states. Although UCCSDSinglet imposes additional parameter-tying constraints, both ansatz span the same excitation manifold, and therefore yield comparable errors. UCCGSD, which provides the most flexible excitation manifold considered here, generally produces the lowest errors, albeit at the cost of considerably greater circuit depth. These results demonstrate the expected trade-off between reference-state expressiveness and circuit complexity.

Furthermore, we note that the influence of the reference ansatz is considerably more pronounced for the ground state than for the excited states. For S0, the difference in median error between the paired and unpaired ansatz exceeds an order of magnitude, whereas the corresponding variation for S1, T1 and T2 is substantially smaller. This behaviour can be understood from the construction of QSE. Although the projected Hamiltonian variationally refines all states within the QSE subspace, the ground-state solution remains strongly dominated by the original VQE reference. Consequently, deficiencies in the reference wavefunction can only be partially corrected through the QSE expansion, making the accuracy of the ground-state energy more sensitive to the quality of the underlying VQE reference. In contrast, the excited states contain more substantial contributions from the other operator-generated basis states, which provide greater flexibility beyond the original reference. These contributions can partially compensate for imperfections in the VQE wavefunction, thereby reducing the sensitivity of the excited-state energies to the choice of ansatz.

Beyond the dependence on the reference ansatz, the excited states are generally recovered with appreciably larger errors than the ground state. Among the low-lying excited states considered, S1 is consistently the least accurately reproduced, while T1 and T2 perform better but generally remain outside chemical accuracy. The errors are also distributed unevenly across the molecular benchmark, with a broad spread and a small number of large-error outliers forming pronounced tails in the distribution (Fig. S1). This skewness motivates the use of the median as a more representative measure of typical performance than the mean.

Root assignment

We next examine the large-error cases in greater detail. We observe that in several instances, the target exact state is not recovered as a distinct state within the QSE spectrum. Pyrrole (UCCSD) provides a clear example of this. The first excited singlet obtained from QSE has an energy of -205.883672Ha, whereas the exact S1 and S2 energies are -205.988082Ha and -205.883871Ha respectively. The QSE root also has negligible overlap with the exact S1 state, 4.14×10^{-6} , but an overlap of 0.997 with S2. The recovered state is therefore an accurate approximation to S2 rather than S1, while no corresponding QSE root is recovered for S1. This corresponds to a different failure mode compared to an exact state that is simply poorly represented.

To better quantify the effect of this index-based root assignment errors, we additionally performed an overlap-based Hungarian matching that maximises the total wavefunction overlap between QSE and exact CASCI states. This removes the effects of missing or wrongly assigned roots on the raw errors, and instead present them as unrecovered states. For UCCSD at 6e6o, S0 and T1 retain complete matching with no change in their errors. For S1 and T2, the fractions of recovered states decrease to 75% and 93%, respectively, while the median errors decrease from 23.7 to 18.2 mHa for S1 and from 3.2 to 2.6 mHa for T2. Therefore, while this additional failure mode can also affect the low-lying states, it does not appear to be the dominant source of the observed errors.

Spin purity

We next examine the spin purity of the recovered QSE spectrum. Although the QSE operator pools used in this study are spin adapted, this does not by itself guarantee spin-pure Ritz states because the underlying VQE reference is not perfectly spin pure. Consequently, small spin-contaminated components of the reference state can be carried into the QSE subspace by the excitation operators. We therefore first examine how these spin-contaminated states are distributed across the recovered spectrum (Fig. S1).

We observe that spin contamination is relatively uncommon among the lower-lying roots, but becomes increasingly prevalent at higher energies. Its extent also depends strongly on the reference ansatz: the paired ansätze produce almost entirely spin-pure spectra, whereas the more expressive UCCGSD ansatz exhibits the most severe spin contamination among the ansätze considered.

Interestingly, the spin-adapted UCCSDSinglet ansatz, which ties parameters associated with equivalent spatial-orbital transitions, exhibits greater spin contamination than the more flexible UCCSD ansatz, both in the number of contaminated roots (Fig. S1) and in the mean spin deviation (Table S1). We attribute this behaviour to errors introduced by the Trotter approximation. As the number of Trotter steps is increased, the spin contamination of UCCSDSinglet decreases systematically, whereas that of UCCSD remains approximately unchanged. In our calculations, the benefit of parameter tying becomes apparent only beyond 16 Trotter steps (Table S2). The greater variational flexibility of UCCSD may therefore make it less sensitive to these Trotterisation errors at low Trotter order, partially offsetting the loss of explicit spin adaptation.

Taken together with the spectral-error analysis above, these results highlight UCCSD as a favourable compromise among the ansätze considered. It maintains relatively low spectral errors while exhibiting substantially less spin contamination than the more expressive UCCGSD ansatz. UCCSD therefore provides a useful balance between spectral accuracy, spin purity, and ansatz complexity within the present implementation.

Geometric representability of exact states

Beyond the failure modes discussed above, we next examine the geometric representability of the target states within the QSE subspace. To quantify this, we define a projection metric for each exact state as the squared norm of its projection onto the full one-body-generated QSE subspace. This directly measures how much of the exact wavefunction is captured by the QSE subspace.

Figure 1 compares the exact-state projection with the corresponding QSE energy error for S0, S1, T1 and T2. For S0, the projections are strongly concentrated near unity, consistent with the ground state being well represented within the QSE subspace. On the other hand, the excited states show reduced projections and larger energy errors, most significantly observed for S1.

As expected, low projection is generally associated with large energy errors, since a substantial component of the exact state lies outside the QSE subspace and cannot be represented by any linear combination of its

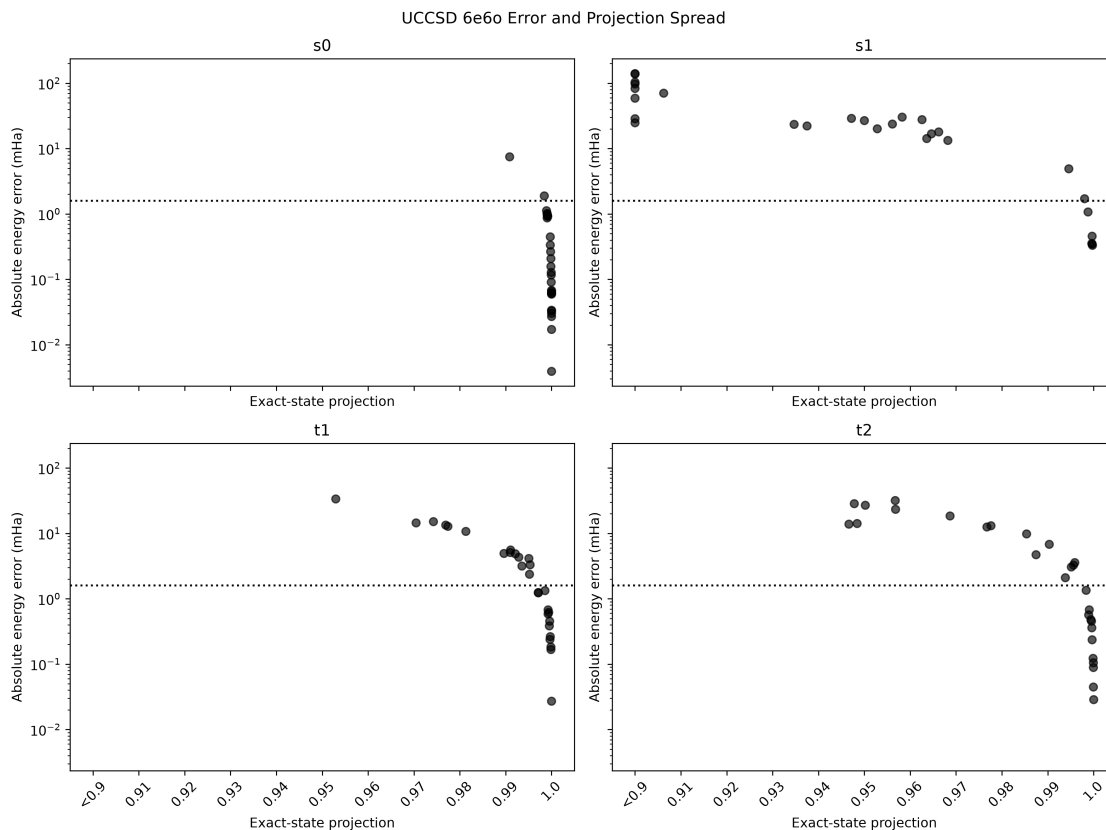


Figure 1: Absolute QSE energy errors as a function of the projection of exact states onto the QSE subspace for UCCSD in the 6e6o active space. Results are shown for the lowest singlet and triplet states, with the dotted line indicating the chemical-accuracy threshold.

basis states. However, the converse is not necessarily true: projections exceeding 0.99 do not consistently yield energies within chemical accuracy, and a considerable spread in error remains even among these highly projected states.

This distinction is important in the context of the original QSE construction, which was motivated by the expectation that low-lying molecular excited states are dominated by single-electron promotions, such that a one-body-generated subspace captures a significant fraction of their wavefunction support [9]. However, the projection results show that capturing the dominant support is not sufficient to guarantee spectral accuracy. Even when an exact state lies almost entirely within the QSE subspace, the remaining missing component can contain energetically important directions that are absent from the projected problem. Moreover, even a small component missing from the QSE subspace can change the orientation of the best Ritz approximation, effectively rotating the recovered state away from the exact target. Thus, a one-body-generated QSE subspace may capture most of the wavefunction while remaining incomplete in the directions required for accurate spectral recovery.

Active-space dependence of representability and accuracy

Lastly, we consider the effects of active space scaling on the accuracy of the recovered QSE spectra (Fig 2). For the smallest active spaces, 2e2o, 2e3o, and 4e3o, the recovered spectra are nearly exact. Beyond these cases, the errors generally increase as the active space is enlarged.

The near-exact recovery in the smallest active spaces is consistent with the relative dimensionality of the QSE subspace. In these cases, the one-body operator pool generates more basis states than there are physically accessible determinants within the relevant spin sector. The QSE subspace can therefore span

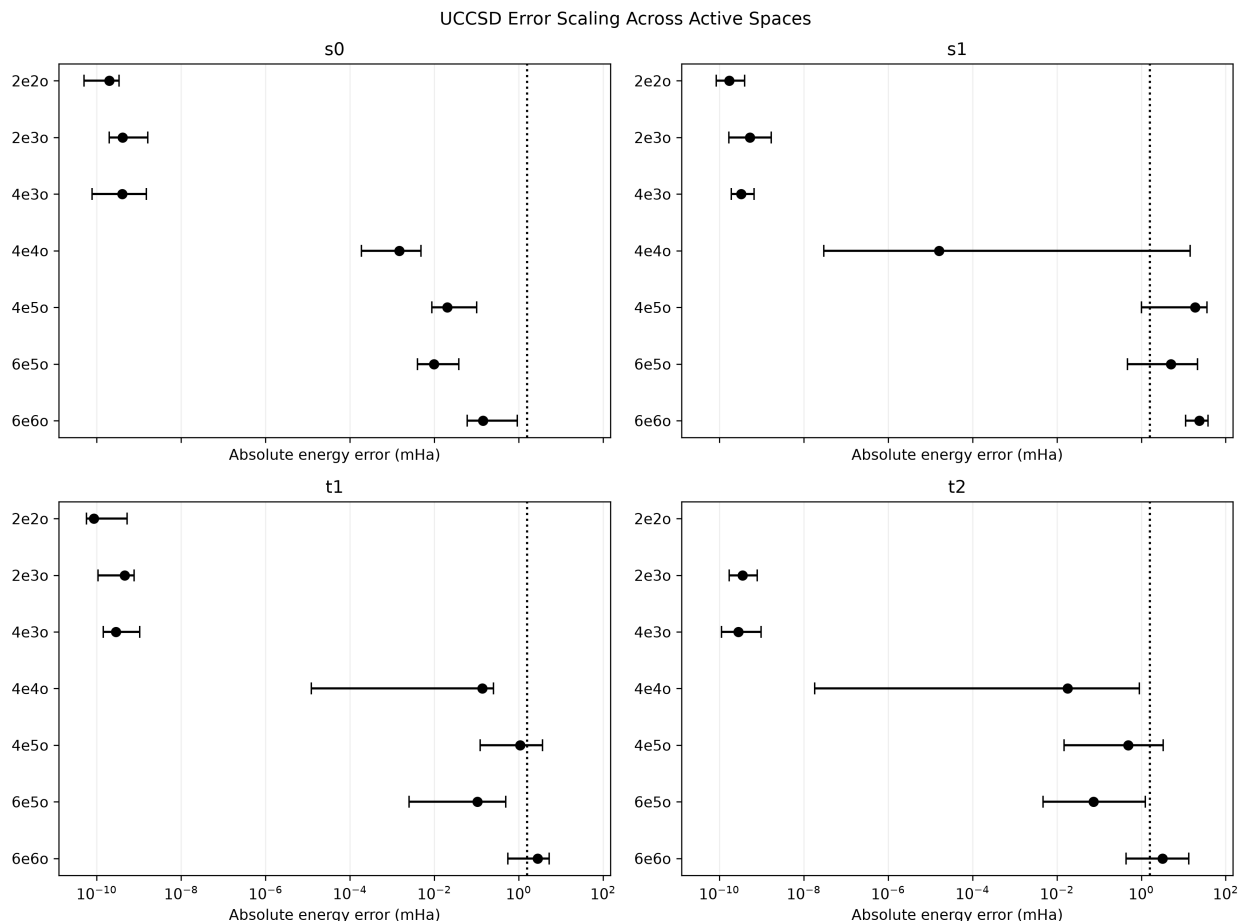


Figure 2: Dependence of UCCSD QSE energy errors on active-space size. Results are shown for the lowest singlet and triplet states across the 2e20, 2e30, 4e30, 4e40, 4e50, 6e50, and 6e60 active spaces. Markers and horizontal error bars summarize the central error and spread across the molecular benchmark, while the dotted line denotes chemical accuracy.

most of the relevant Hilbert space, leaving little room for representability error.

This favourable dimensionality does not persist as the active space grows. The number of one-body QSE operators scales polynomially, approximately as n^2 , whereas the dimension of the full configuration-interaction (FCI) space grows combinatorially with the number of electrons and orbitals. Therefore, the fraction of the many-electron space accessible to the generated QSE subspace decreases with system size.

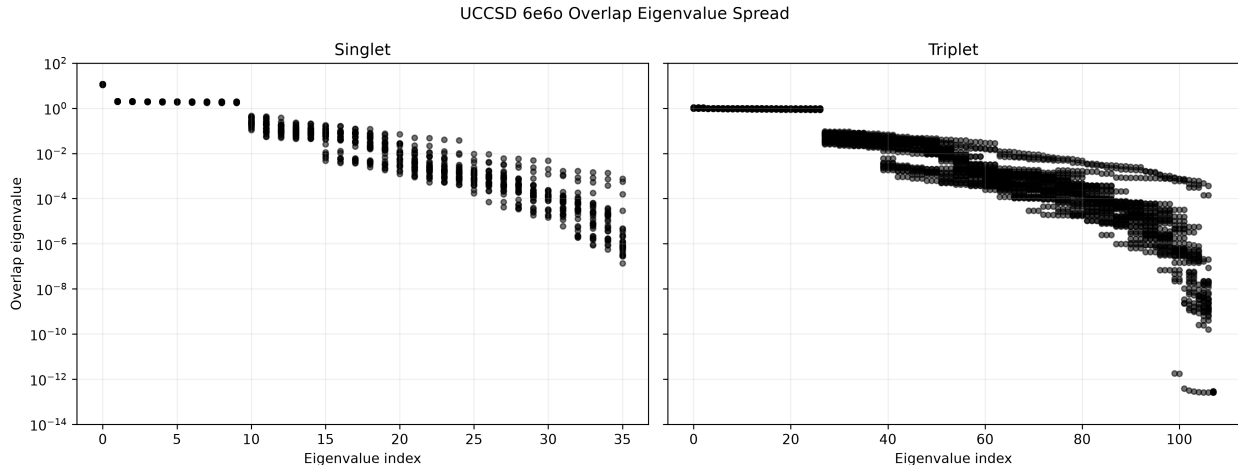
At the largest active spaces considered in this study, the low-lying excited states are still not recovered to chemical accuracy. This suggests that one-body expansion operators alone may be insufficient for chemically accurate excited-state calculations, and that a larger operator manifold consisting of two-body excitation operators may instead be required. Consequently, determining how the choice and size of the operator pool affect representability and spectral accuracy as active space grows will be essential for assessing the practical applicability of QSE beyond the relatively small active spaces considered in this work.

* TBC 6e70, 8e70 and 8e80 data will include here later

Stable Dimensions of QSE

Eigenvalue Distribution

(a)



(b)

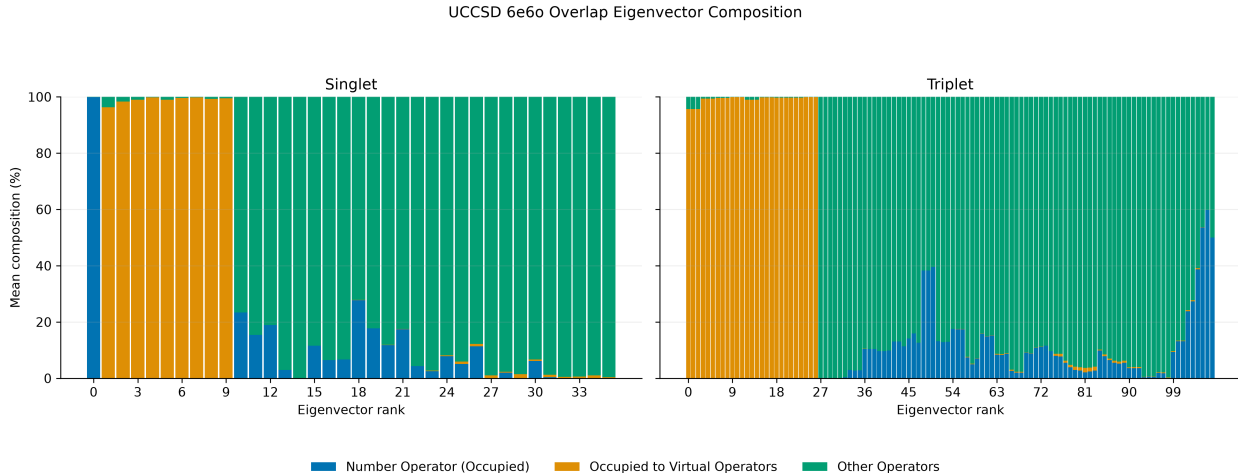


Figure 3: Structure of the UCCSD overlap matrix in the 6e6o active space. The upper panel shows the overlap-eigenvalue spectrum for the singlet and triplet sectors across the molecular benchmark. The lower panel decomposes the corresponding overlap eigenvectors into contributions from occupied-number operators, occupied-to-virtual operators, and other operators.

Having examined the representability and spectral accuracy of the QSE subspace, we next turn to its numerical stability. As discussed in the Introduction, very small eigenvalues of the overlap matrix S can strongly amplify noise, making the generalised eigenvalue problem highly sensitive to perturbations. Here, we examine the full eigenvalue spectrum of S (Figure 2(a)) to understand why these near-null directions arise in the first place.

One clear trend that emerges is between the eigenvalue magnitude and the composition of its corresponding eigenvector. We first classify the QSE excitation operators into three groups: occupied number operators, occupied-to-virtual (OV) excitations, and the remaining excitation classes. Each eigenvector of S can then be decomposed into basis states corresponding to each category (Figure 2(b)). One clear trend emerges between the eigenvalue magnitude and the composition of its corresponding eigenvector. The largest eigenvalues are generally associated with OV-dominated eigenmodes. Singlets additionally contain a strongly supported

occupied-number mode, while the remaining operator classes contribute predominantly to smaller-eigenvalue directions. For the singlet subspace, this produces one number-dominated eigenvector followed by a block of OV-dominated eigenvectors with eigenvalues close to unity, after which the spectrum drops to progressively smaller values. The triplet subspace retains the OV-dominated stable block, but the corresponding occupied-number contribution is much more weakly supported.

This behaviour can be understood from how the different QSE operators act on the underlying VQE reference. Because the reference state is dominated by the Hartree Fock determinant, operators that act directly on this component generate strongly supported basis directions and therefore larger overlap eigenvalues. In contrast, operators that annihilate the dominant Hartree Fock component, or act primarily on the smaller correlated components of the reference state, generate much weaker directions and correspondingly smaller eigenvalues.

This operator-support picture also explains the contrasting behaviour of the occupied number operators in the singlet and triplet sectors. As defined in eqs. (12) and (15), the diagonal singlet operator for an occupied spatial orbital i is proportional to $\hat{n}_{i\alpha} + \hat{n}_{i\beta}$, so that

$$\hat{O}_{ii}^S |\text{HF}\rangle = (\hat{n}_{i\alpha} + \hat{n}_{i\beta}) |\text{HF}\rangle = 2 |\text{HF}\rangle. \quad (4)$$

The dominant Hartree-Fock component therefore provides strong support for this direction, consistent with the large singlet overlap eigenvalues observed. In contrast, the $M_S = 0$ triplet diagonal operator is proportional to $\hat{n}_{i\alpha} - \hat{n}_{i\beta}$, which gives

$$\hat{O}_{ii}^{T,0} |\text{HF}\rangle = (\hat{n}_{i\alpha} - \hat{n}_{i\beta}) |\text{HF}\rangle = 0. \quad (5)$$

The $M_S = \pm 1$ diagonal triplet operators likewise annihilate the closed-shell Hartree-Fock determinant because the corresponding spin-flip attempts to create an electron in an already occupied spin orbital. The triplet number operators therefore receive support only from configurations containing singly occupied spatial orbitals. This is particularly consequential for 1-UpCCGSD, which only allows for singles and paired doubles excitations. Its spectrum shown in Fig. S3, highlights the corresponding M number-operator directions appearing as near-null eigenvalues.

A second source of near-null directions observed occurs within the $2e2o$, $2e3o$, and $4e3o$ active spaces. In those cases, the number of nominal QSE basis states exceeds the dimension of the physically accessible subspace within the relevant spin sector. This results in redundant linear combinations, which manifest as near null eigenvalues of the overlap matrix S . More generally, this occurs for small active spaces in which the orbital window does not extend beyond $\text{HOMO} \pm 1$ and $\text{LUMO} \pm 1$. The resulting near-null eigenvalues are therefore not caused by weak operator support alone, but can arise simply from the dimensionality constraint of the many-electron Hilbert space.

Significance of Small Eigenvalues

Taken together, these trends reveal a recurring block of comparatively large overlap eigenvalues associated with the strongly supported operator classes identified above. The corresponding stable dimensions are therefore approximately $1 + n_{\text{occ}}n_{\text{vir}}$ for singlets and $3n_{\text{occ}}n_{\text{vir}}$ for triplets, with the remaining directions progressively less well supported. We term these dimensions *stable* because their associated overlap eigenvalues remain consistently of order 1 or larger. Retaining these directions therefore does not introduce any strong noise amplifications.

This naturally raises the question of whether these *stable* dimensions alone are sufficient to recover most of the low-energy QSE spectrum. This is of significant interest because a consistently stable rank would provide a natural basis for thresholding, particularly when shot-sampling noise is introduced. To test this, we restrict the QSE subspace to the stable, strongly supported dimensions identified above and solve the resulting generalised eigenvalue problem. However, restricting the QSE subspace to only these stable dimensions severely degrades the recovered spectrum, producing markedly larger energy errors across the target states (Fig. S4).

This suggests that while the stable dimensions form a well-conditioned core of the QSE subspace, they themselves are insufficiently to recover the low-lying states accurately. The remaining weakly supported directions contribute important correlation effects that cannot be trivially discarded. This establishes a

central trade-off: removing weak directions improves numerical stability, but can simultaneously degrade the representability and accuracy of the QSE subspace. The next question is therefore how to choose the retained overlap eigenmodes so as to balance numerical stability against the representability and spectral accuracy of the QSE subspace in the presence of noise.

Finite-Shot Effects on Generalized Eigenvalue Stability

Effects of Thresholding to mitigate Sampling Noise

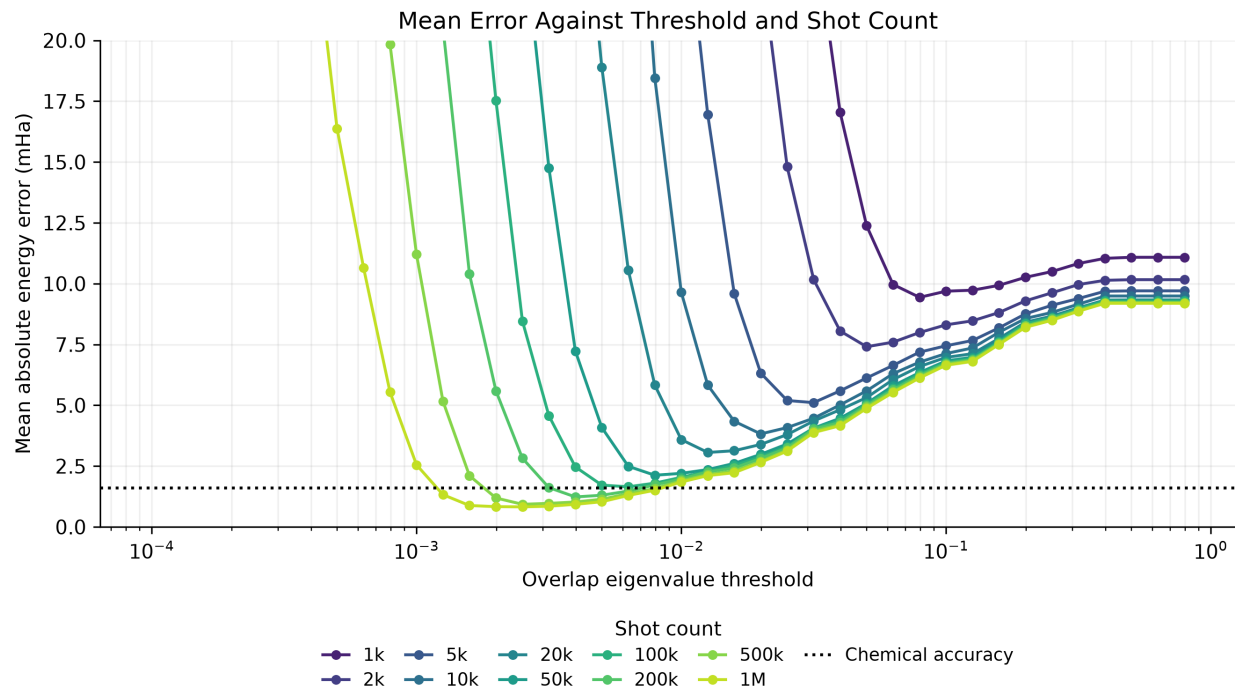


Figure 4: Mean absolute QSE energy error as a function of the overlap-eigenvalue threshold for the *6e6o* active space at different shot counts. The dotted line indicates chemical accuracy, and the curves show the trade-off between suppressing sampling-noise amplification and discarding physically relevant subspace directions.

Active Space Dependence of Shot Noise

Shot Budget Scaling

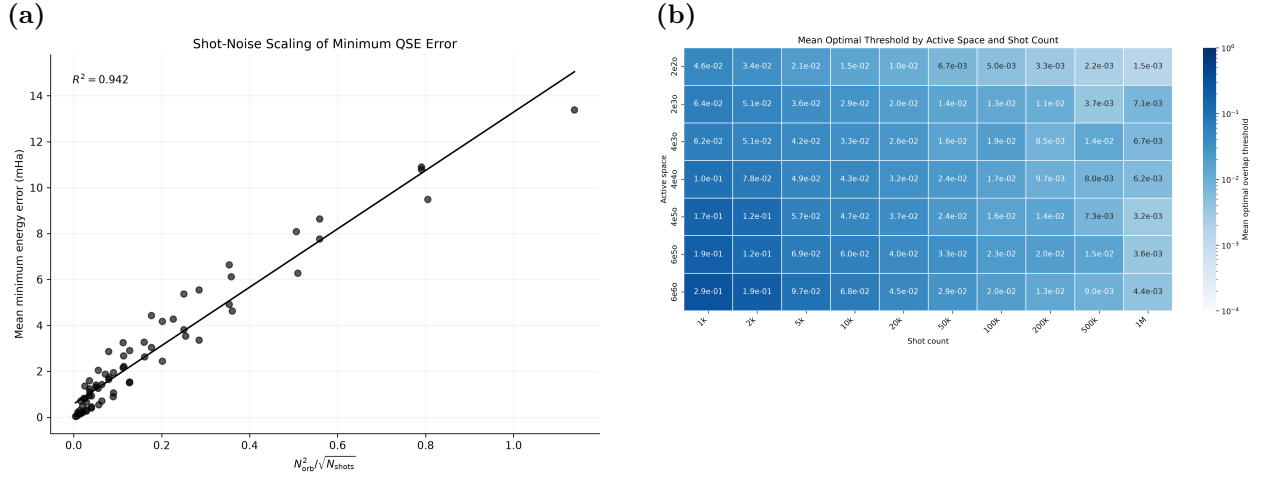


Figure 5: Finite-shot scaling and threshold selection across active spaces. (a) Minimum mean QSE energy error as a function of the active-space scaling variable $N_{\text{orb}}^2 / \sqrt{N_{\text{shots}}}$, with the linear regression and coefficient of determination shown. (b) Mean optimal overlap-eigenvalue threshold as a function of active space and shot count.

Structure of the Projected QSE Matrices

Structural Correlations and Sparsity

Effects of Noise on Structure

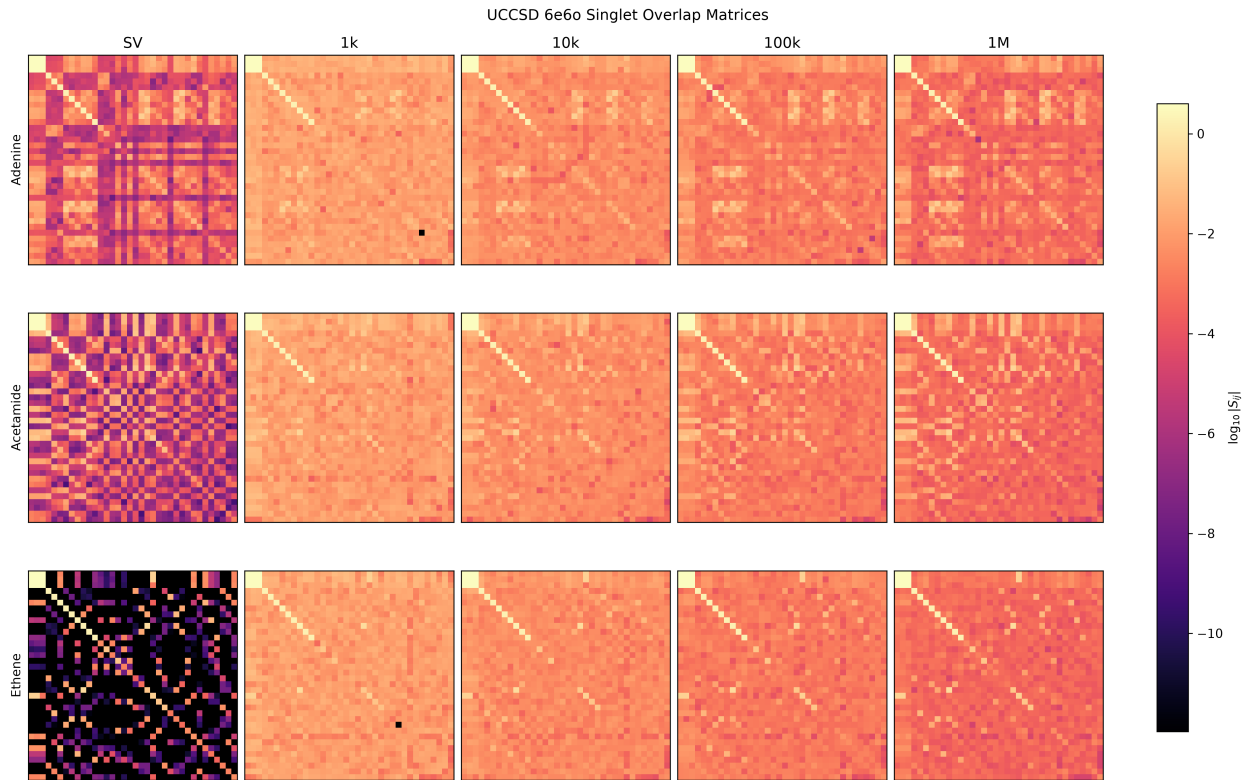


Figure 6: Effect of finite-shot noise on the structure of UCCSD overlap matrices in the 6e6o singlet sector. The statevector matrix is compared with matrices reconstructed using shot counts from 1 k to 1 M, illustrating the progressive disruption of the underlying matrix structure by sampling noise.

Outlook for Structure-Aware reconstruction

Computational Methods

Molecular Geometries and Active Space Preparation

The dataset used comprised 28 closed-shell organic molecules from Thiel’s excited-state benchmark set [13]. The published Cartesian nuclear coordinates were referenced directly and used without further optimisation. For each molecule, Restricted Hartree-Fock calculations were performed using PySCF [14, 15] with the STO-3G basis set. This set of canonical molecular orbitals was optimised once and used consistently throughout all subsequent CASCI, VQE and QSE calculations in the study.

Active spaces (denoted $n_e n_o$) comprised the $n_e/2$ occupied molecular spatial-orbitals closest to the HOMO and the $n_o - n_e/2$ virtual molecular spatial-orbitals closest to the LUMO. The remaining occupied orbitals were treated as frozen and doubly occupied, while the remaining virtual orbitals were excluded from calculations.

Exact Reference

Within each active space, the exact reference energies and wavefunctions were obtained from complete active-space configuration interaction (CASCI) calculations performed using PySCF. Calculations were separated by spin-projection sector M_S by setting the number of active α and β spin electrons to be

$$n_\alpha = \frac{n_e}{2} + M_S, n_\beta = \frac{n_e}{2} - M_S$$

Singlet states were obtained from the $M_S = 0$ sector, while triplet states were calculated in the $M_S = 0, +1, -1$ sectors. For each active-space spin projection sector, the determinant-space dimension, and therefore the maximum number of available roots, was $\binom{n_o}{n_\alpha} \binom{n_o}{n_\beta}$.

Where available, the lowest 11 singlet states (ground state + 10 excited states), and the lowest 10 triplet states (within each spin projection sector) were calculated. To ensure that the target number of states was obtained, three times the required number of CASCI roots were initially requested from the FCI solver, subject to the maximum dimension of the corresponding determinant space. The resulting roots were then filtered according to their $\langle \hat{S}^2 \rangle$ expectation values, retaining only roots that satisfy $|\langle \hat{S}^2 \rangle - S(S+1)| < 10^{-5}$ with $S = 0$ for singlets and $S = 1$ for triplets.

Throughout this study, wavefunctions were represented in a common active spin-orbital determinant basis by a state vector of length $\binom{2n_o}{n_e}$. This vector includes all determinants with n_e electrons distributed among the $2n_o$ active spin-orbitals (irrespective of their spin projections). Determinants were ordered lexicographically using an interleaved spin-orbital convention ($1\alpha, 1\beta, 2\alpha, 2\beta, \dots$).

PySCF represented each wavefunction as a two-dimensional CI coefficient matrix with dimensions $\binom{n_o}{n_\alpha}$ by $\binom{n_o}{n_\beta}$, corresponding to the α and β electron occupation strings. These coefficients were embedded into the corresponding entries of the common state-vector representation. During this conversion, the fermionic phase arising from the permutation between PySCF’s separated α and β electron ordering and the interleaved spin-orbital convention was included explicitly.

Quantum Computing Framework

All quantum-circuit calculations in this study were performed using classical simulations implemented with the Qibo and Qibochem [16, 17] frameworks. These frameworks were used to construct, represent and simulate the quantum circuits employed. A research fork of Qibochem was developed to support the computational workflow used in the study. The fork introduced configurable ansatz classes, optimised circuit evaluation routines, support for measuring spin observables, and dedicated functionality for constructing and sampling the projected matrices required for Quantum Subspace Expansion calculations.

All calculations in this work used ideal noiseless circuit simulations without modelling any hardware. Only finite-sampling noise arising from a limited number of measurement shots was considered in the relevant sections.

Hamiltonian and Spin Operators

For each active space, the electronic Hamiltonian was constructed in second quantisation as follows:

$$\hat{H} = E_{\text{nuc}} + \sum_{pq} h_{pq} \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} \sum_{pqrs} g_{pqrs} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r. \quad (6)$$

where E_{nuc} is the core nuclear repulsion energy and p, q, r, s index the active spin orbitals. h_{pq} and g_{pqrs} are the one and two-electron integral coefficients evaluated in the fixed canonical Hartree-Fock MO basis.

Spin operators were constructed in the same second quantisation basis as follows.

$$\hat{S}_z = \frac{1}{2} \sum_p (\hat{n}_{p\alpha} - \hat{n}_{p\beta}), \quad \hat{n}_{p\sigma} = \hat{a}_{p\sigma}^\dagger \hat{a}_{p\sigma}. \quad (7)$$

$$\hat{S}_+ = \sum_p \hat{a}_{p\alpha}^\dagger \hat{a}_{p\beta}, \quad \hat{S}_- = \sum_p \hat{a}_{p\beta}^\dagger \hat{a}_{p\alpha}. \quad (8)$$

$$\hat{S}^2 = \hat{S}_z^2 + \frac{1}{2} (\hat{S}_+ \hat{S}_- + \hat{S}_- \hat{S}_+). \quad (9)$$

All fermionic Hamiltonian, spin and excitation operators were mapped to qubit operators using the Jordan-Wigner (JW) transformation.

Ground State Reference

As introduced above, approximate ground-state reference wavefunctions were prepared using parameterised Unitary Coupled-Cluster (UCC) Ansatz. In this study, six ansatz families were used, namely: UCCSD, UCCSDSinglet, UCCGSD, and k -UpCCGSDSinglet with $k = 1, 2$, and 3. The Ansatz were characterised by the excitation generators included in the cluster operator $\hat{T}$, summarised in the following table.

Table 2: Classification of the 6 UCC ansatz families

Ansatz	Singles excitations	Doubles excitations	Spin- conserving	Generalised	Spin- adapted	Paired
UCCSD	✓	✓	✓			
UCCSDSinglet	✓	✓	✓		✓	
UCCGSD	✓	✓	✓	✓		
1UpCCGSDSinglet	✓	✓	✓	✓	✓	✓
2UpCCGSDSinglet	✓	✓	✓	✓	✓	✓
3UpCCGSDSinglet	✓	✓	✓	✓	✓	✓

For each ansatz, candidate singles and doubles transitions were first enumerated in fermionic form. Only canonical transitions that excite from a lower to higher spin orbital were retained because the reverse transitions would already be included through $\hat{T}^\dagger$ before exponentiation. All retained generators were spin conserving: single excitations preserved spin labels, while double excitations preserved the total number of α and β spin electrons.

The non-generalised UCCSD and UCCSDSinglet ansatz were restricted to occupied-to-virtual single and double excitations relative to the reference Hartree-Fock determinant [18]. In contrast, the generalised UCCGSD ansatz allowed generalised orbital transitions between occupied-occupied and virtual-virtual spin orbitals [19, 20]. To reduce the size of the generalised-double operator pool, mixed-direction double excitations, in which one electron was promoted while the other was relaxed, were excluded. Across the *4e4o* to *8e8o* active spaces, this practical restriction reduced the number of independent parameters by 42.2–45.2% and the circuit depth by 46.4–47.6%.

The k -UpCCGSDSinglet ansatz combined generalised singles with generalised paired doubles. For each paired double excitation, an $\alpha\beta$ electron pair was removed from one spatial orbital and created in another. The UCCSDSinglet and k -UpCCGSDSinglet constructions imposed spin-adapted parameter sharing between spin-orbital excitations that had the same spatial-orbital labels. For k -UpCCGSDSinglet, the complete

generalised single and paired-double block was repeated k times, with an independent set of variational parameters assigned to each repetition.

Unless otherwise stated, all UCC operators were implemented using the first-order Suzuki Trotter product approximation given in eq. (10).

$$e^{\hat{T}-\hat{T}^\dagger} \approx \prod_{\lambda} e^{\theta_{\lambda}(\hat{T}_{\lambda}-\hat{T}_{\lambda}^\dagger)}. \quad (10)$$

At every optimisation step, the circuit energy was evaluated as the noiseless statevector expectation value of the electronic Hamiltonian eq. (6). The variational parameters were optimised classically using the L-BFGS-B optimiser through Qibo, to minimise the ground state expectation value. For the k -UpCCGSDSinglet ansatz, the parameters associated with all k repeated layers were optimised simultaneously. Before optimisation, ansatz parameters were initialised based on the following hierarchy:

$$\text{MP2} \rightarrow \text{1-UpCCGSD} \rightarrow \text{2-UpCCGSD} \rightarrow \text{3-UpCCGSD} \rightarrow \text{UCCSDSinglet} \rightarrow \text{UCCSD} \rightarrow \text{UCCGSD}.$$

Parameters associated with excitation generators shared with the preceding ansatz were initialised with the previously optimised amplitudes, while newly introduced generators were initialised to zero. The initial occupied-to-virtual double-excitation parameters were estimated using second-order Moller-Plesset perturbation theory (MP2). For parameter transfer from a k -UpCCGSDSinglet ansatz to a subsequent ansatz, the optimized amplitudes associated with each generator were summed across the k layers and used as the initial amplitude for the corresponding generator.

Following optimisation, the resulting VQE parameters were stored as a fixed ground-state reference wavefunction. This fixed reference set was used consistently for all subsequent statevector and finite-shot QSE calculations.

QSE Operator Pools and Projected Matrices

As discussed in the introduction, separate one-body operator pools were used to construct the singlet and triplet-targeting QSE subspaces. The singlet operator pool, $\mathcal{O}^S$, was defined as

$$\hat{O}_{ij}^S = \hat{a}_{i\alpha}^\dagger \hat{a}_{j\alpha} + \hat{a}_{i\beta}^\dagger \hat{a}_{j\beta}, \quad (11)$$

$$\mathcal{O}^S = \left\{ \hat{O}_{ij}^S \mid i, j \in \{1, \dots, n_o\} \right\}. \quad (12)$$

whereby i and j to denote active spatial orbitals.

The triplet operator pool, $\mathcal{O}^T$, was constructed from all three spin-projection components, which were collected into a single operator pool.

$$\hat{O}_{ij}^{T,0} = \frac{1}{\sqrt{2}} \left(\hat{a}_{i\alpha}^\dagger \hat{a}_{j\alpha} - \hat{a}_{i\beta}^\dagger \hat{a}_{j\beta} \right), \quad (13)$$

$$\hat{O}_{ij}^{T,+1} = \hat{a}_{i\alpha}^\dagger \hat{a}_{j\beta}, \quad \hat{O}_{ij}^{T,-1} = \hat{a}_{i\beta}^\dagger \hat{a}_{j\alpha}, \quad (14)$$

$$\mathcal{O}^T = \left\{ \hat{O}_{ij}^{T,m} \mid i, j \in \{1, \dots, n_o\}, m \in \{-1, 0, +1\} \right\}. \quad (15)$$

The singlet operator pool therefore contains n_o^2 operators, while the triplet pool contains $3n_o^2$ operators. Individual pool elements were enumerated as $\{\hat{O}_\mu\}$, with μ and ν indexing arbitrary operators within the given pool.

A projected observable matrix were constructed for the overlap, Hamiltonian and spin-squared matrices. Observable matrix elements were defined as follows:

$$\bar{s}_{\mu\nu} = \hat{O}_\mu^\dagger \hat{O}_\nu, \quad \bar{h}_{\mu\nu} = \hat{O}_\mu^\dagger \hat{H} \hat{O}_\nu, \quad \bar{z}_{\mu\nu} = \hat{O}_\mu^\dagger \hat{S}^2 \hat{O}_\nu. \quad (16)$$

where the corresponding target observables were $\hat{I}$, $\hat{H}$, and $\hat{S}^2$, respectively.

The fermionic QSE expansion operators and each target observable were mapped separately to Pauli operators using the Jordan-Wigner transformation. The projected observables were then constructed by

multiplication in the Pauli representation. Terms with coefficients of magnitude smaller than 10^{-12} were discarded after the final product construction. The resulting Pauli-string expectation values were then evaluated with respect to the optimised VQE states, $|\Psi_{\text{VQE}}\rangle$. For all calculations performed, only the diagonal and upper-triangular matrix elements, $\mu \leq \nu$ were constructed and evaluated. Lower-triangular matrix terms were then reconstructed using Hermitian symmetry.

Finite-Shot Sampling

Finite-shot sampling was used to model the effects of statistical measurement noise on the projected Hamiltonian and overlap matrices. For each calculation, all unique Pauli strings in the projected Hamiltonian and overlap observable matrices were first collected into a global measurement pool. Then, a largest-first greedy qubit-wise-commuting (QWC) grouping algorithm was used to partition the global measurement pool. Grouping was performed globally across elements from both matrices which reduced the total number of commuting groups that needed to be measured. Preliminary data also showed that this method of global grouping, as opposed to grouping terms for each matrix element, resulted in much better agreement with the statevector results.

For each commuting group, a separate measurement circuit was constructed by appending basis-rotation gates to the optimized VQE circuit, such that all Pauli strings in the group could be measured in the computational (Z) basis. Hadamard gates were applied to qubits measured in the (X) basis, while $S^\dagger$ followed by a Hadamard gate was applied to qubits measured in the (Y) basis. This produced a distinct rotated statevector for each commuting group.

Measurement outcomes were generated for each rotated statevector using NumPy's multinomial sampler. The expectation values of all Pauli strings within a commuting group were estimated from the resulting sampled frequencies. These expectation values were then used to reconstruct the Hamiltonian and overlap matrices.

At each specified shot count, the finite-shot calculation was repeated using ten statistically independent sampling iterations. The resulting Hamiltonian and overlap matrices were solved independently, and the reported energy errors were calculated from the mean of the ten independently obtained energies.

Numerical Analysis and Evaluation Metrics

Generalised Eigenvalue Problem

Due to the non-orthogonality of the QSE basis states, solving for the energy eigenvalues and wavefunctions involves solving a generalised eigenvalue problem $Hc = ES c$. This was done by first diagonalising the Hermitian overlap matrix S .

$$\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\dagger. \quad (17)$$

where the columns of $\mathbf{U}$ are the eigenvectors of $\mathbf{S}$ and $\mathbf{\Lambda}$ is a diagonal matrix containing the eigenvalues of $\mathbf{S}$. Then, we define the whitening factor

$$\mathbf{V} = \mathbf{U}\mathbf{\Lambda}^{-1/2}, \quad \mathbf{V}^\dagger \mathbf{S} \mathbf{V} = \mathbf{I}. \quad (18)$$

Using $c = \mathbf{V}\mathbf{y}$, we get

$$\mathbf{V}^\dagger \mathbf{H} \mathbf{V} \mathbf{y} = \mathbf{E} \mathbf{V}^\dagger \mathbf{S} \mathbf{V} \mathbf{y} = \mathbf{E} \mathbf{y}. \quad (19)$$

Numerical instabilities arise when the eigenvalues used are tiny or near-null, because noise perturbations will be greatly amplified by a factor $\frac{1}{\sqrt{\lambda}}$. Therefore in practice, the following reduced eigenvalue equation is solved

$$\tilde{\mathbf{V}}^\dagger \mathbf{H} \tilde{\mathbf{V}} \mathbf{y} = \mathbf{E} \mathbf{y}. \quad (20)$$

where $\tilde{\mathbf{V}}$ and $\tilde{\mathbf{V}}^\dagger$ denote a reduced form of $\mathbf{V}$ and $\mathbf{V}^\dagger$, retaining only a selected subset of its columns. This gives the wavefunction in the QSE operator basis as $\mathbf{c}_k = \tilde{\mathbf{V}} \mathbf{y}_k$.

For all statevector calculations used in this work, the retained subset was chosen from a fixed maximum dimension determined by the active space and spin sector as follows.

$$\begin{aligned}
d_S^{\max} &= \min \left[\binom{n_o}{n_p}^2 - \binom{n_o}{n_p+1} \binom{n_o}{n_p-1}, n_o^2 \right], \\
d_T^{\max} &= \min \left[3 \left(\binom{n_o}{n_p+1} \binom{n_o}{n_p-1} - \binom{n_o}{n_p+2} \binom{n_o}{n_p-2} \right), 3n_o(n_o-1) \right], \quad n_p = \frac{n_e}{2}.
\end{aligned} \tag{21}$$

The first term in each expression gives the maximum number of physically distinct singlet or triplet roots available within the corresponding active space. These values were obtained from differences between the dimensions of adjacent fixed- M_S determinant sectors, such that

$$N_S = D_{M_S=S} - D_{M_S=S+1}$$

This subtraction isolates the number of spin- (S) multiplets because the $(M_S = S)$ sector contains one component from every state with total spin (S) or greater, whereas the $(M_S = S + 1)$ sector contains only components with total spin greater than (S) .

For the $2e2o$, $2e3o$, and $4e3o$ active spaces, the number of physically available roots was smaller than the nominal number of QSE expansion operators. The retained dimension was therefore capped by the corresponding physical root count. This cap was in place because it was noted empirically that additional operator directions beyond the physically available space produced near-null eigenvalues. More generally, this restriction applies to active spaces that do not include orbitals extending from HOMO-1 through LUMO+1.

For the triplet subspace, the retained dimension was further limited to $(3n_o(n_o-1))$. The triplet number operators were extremely weakly supported by the paired reference states and generated overlap eigenvalues that could not be resolved reliably above numerical precision. For consistency across ansatz, these directions were therefore excluded for all triplet statevector calculations.

For finite-shot calculations, a threshold τ was used to define the retained subspace. Only eigenvectors with eigenvalues satisfying $\lambda_i > \tau$ were retained. The thresholds τ were drawn from a fixed logarithmic grid from 10^{-4} to 10^0 in increments of 0.1 in the exponent.

Spin-expectation Evaluation

For the k^{th} recovered state, the spin expectation value was calculated as follows:

$$\langle \hat{S}^2 \rangle_k = \frac{\mathbf{c}_k^\dagger \mathbf{Z} \mathbf{c}_k}{\mathbf{c}_k^\dagger \mathbf{S} \mathbf{c}_k}. \tag{22}$$

Using $\mathbf{c}_k = \tilde{\mathbf{V}} \mathbf{y}_k$ and $\tilde{\mathbf{V}}^\dagger \mathbf{S} \tilde{\mathbf{V}} = \mathbf{I}$, this expression becomes

$$\langle \hat{S}^2 \rangle_k = \frac{\mathbf{y}_k^\dagger \tilde{\mathbf{V}}^\dagger \mathbf{Z} \tilde{\mathbf{V}} \mathbf{y}_k}{\mathbf{y}_k^\dagger \mathbf{y}_k}. \tag{23}$$

Projection

TBC results first

Triplet Degeneracies

TBC results first

Non Index State Matching

TBC results first

Conclusions

Acknowledgements

Data Availability

Supporting Information

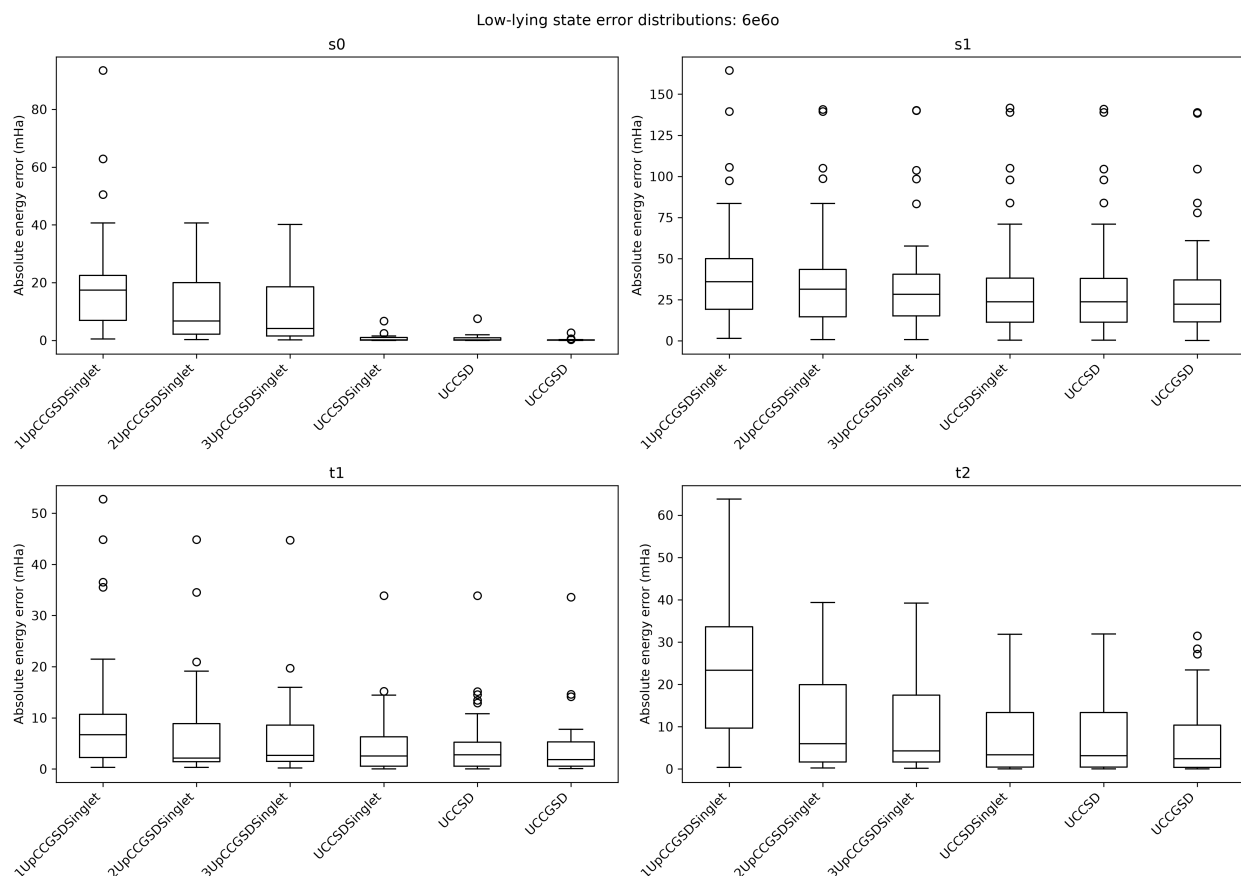


Figure S1: UCCSD low-lying-state error distribution in the 6e6o active space.

Table S1: Mean spin contamination across the molecular benchmark for the low-lying QSE roots in the 6e6o active space.

Ansatz	$s0$	$s1$	$t1$	$t2$
1UpCCGSDSinglet	1.71×10^{-12}	1.93×10^{-10}	4.22×10^{-12}	1.58×10^{-10}
2UpCCGSDSinglet	8.32×10^{-12}	3.86×10^{-10}	2.12×10^{-12}	1.92×10^{-11}
3UpCCGSDSinglet	4.16×10^{-12}	2.34×10^{-10}	2.60×10^{-12}	3.25×10^{-11}
UCCSDSinglet	4.51×10^{-4}	3.23×10^{-4}	1.37×10^{-4}	1.84×10^{-4}
UCCSD	2.31×10^{-4}	1.45×10^{-4}	1.04×10^{-4}	1.13×10^{-4}
UCCGSD	1.93×10^{-4}	3.91×10^{-3}	1.49×10^{-4}	2.14×10^{-4}

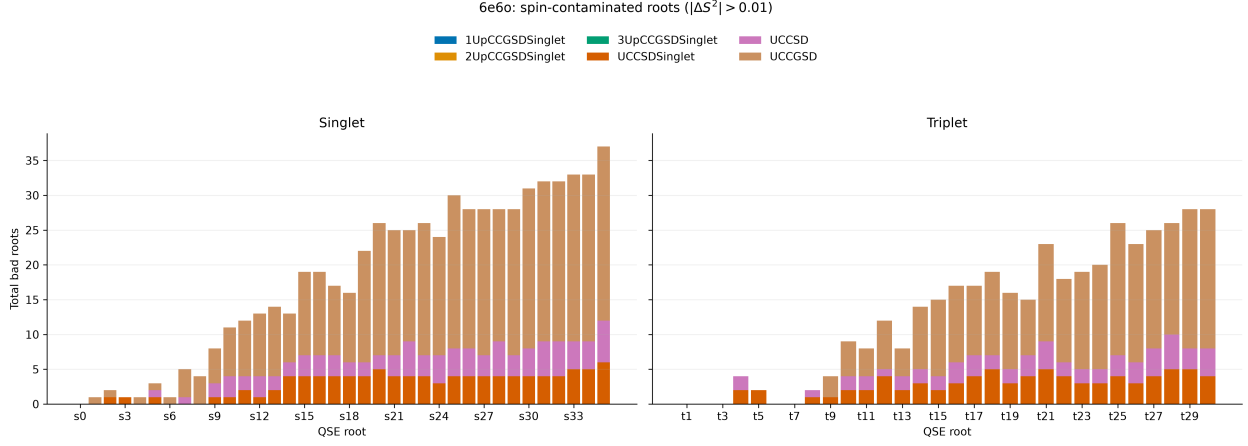


Figure S2: Spin-contaminated roots across ansatz families in the 6e6o active space.

Table S2: Mean spin contamination and circuit depth for the UCCSDSinglet and UCCSD ansatzes across Trotter steps in the 4e4o active space.

Trotter steps	UCCSDSinglet		UCCSD	
	Mean error	Circuit depth	Mean error	Circuit depth
1	2.94×10^{-6}	2181	1.21×10^{-8}	1951
2	6.89×10^{-7}	4370	1.37×10^{-8}	3939
4	1.69×10^{-7}	8748	2.08×10^{-8}	7915
8	4.21×10^{-8}	17 504	1.93×10^{-8}	15 867
16	1.05×10^{-8}	35 016	1.67×10^{-8}	31 771
32	2.62×10^{-9}	70 040	1.65×10^{-8}	63 579

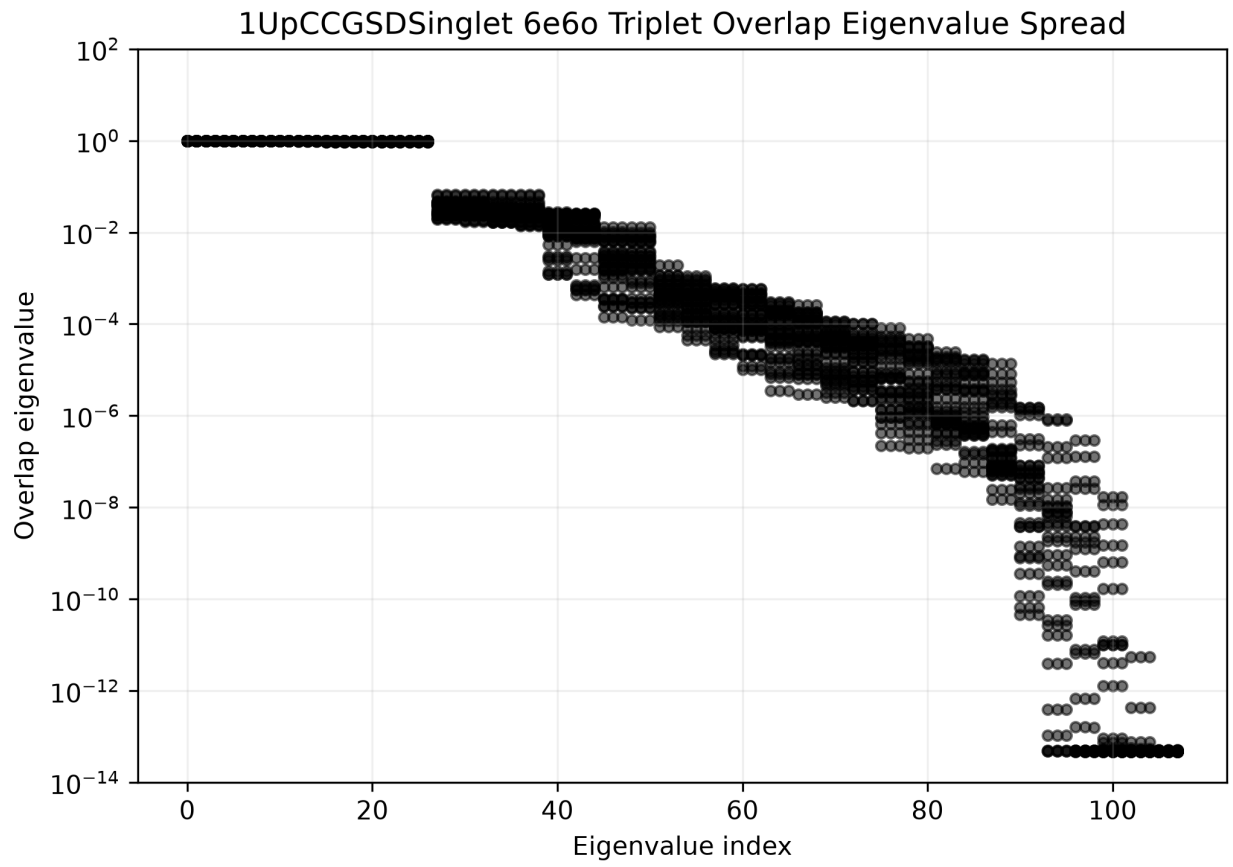


Figure S3: 1UpCCGSDSinglet overlap-eigenvalue spread for the 6e6o singlet and triplet sectors.

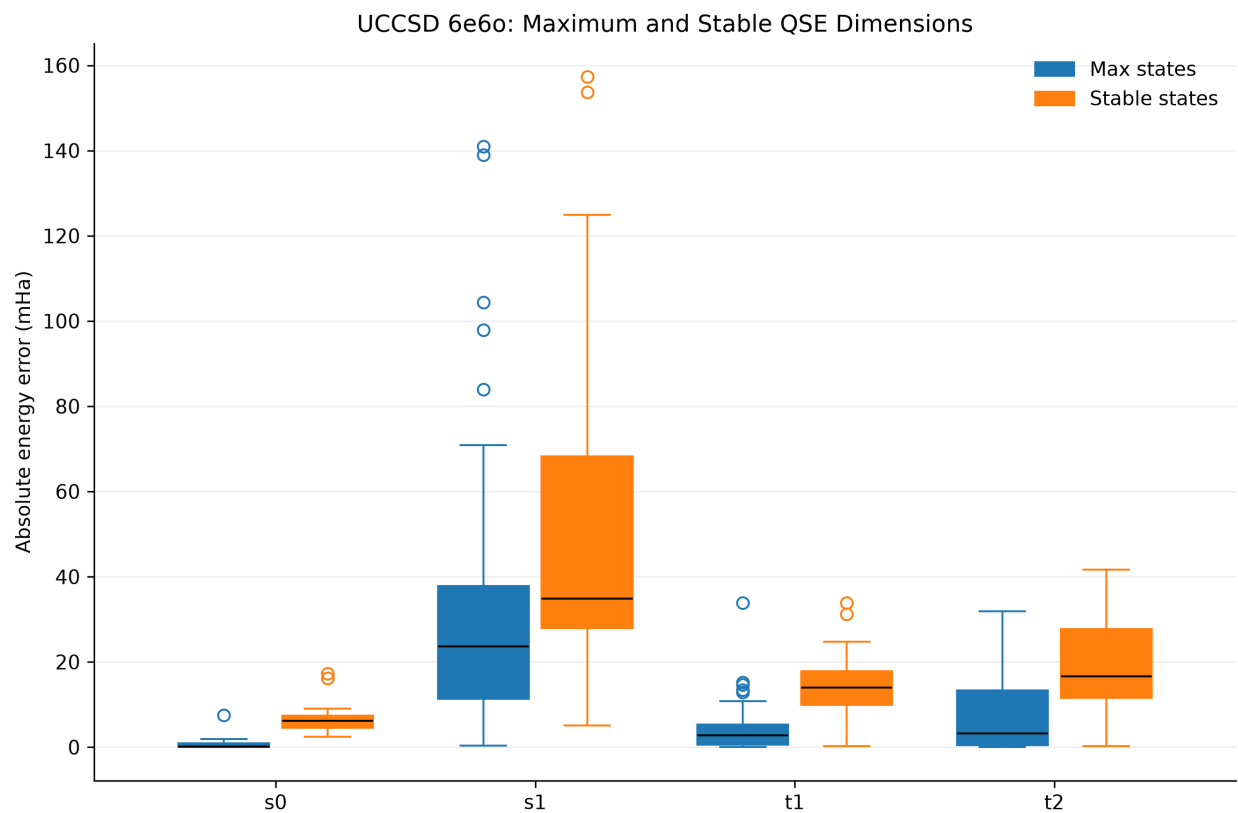


Figure S4: Molecule-level index-matched errors for the low-lying s_0 , s_1 , t_1 , and t_2 roots in the UCCSD 6e6o active space, comparing the maximum retained QSE dimension with the overlap-stable dimension.

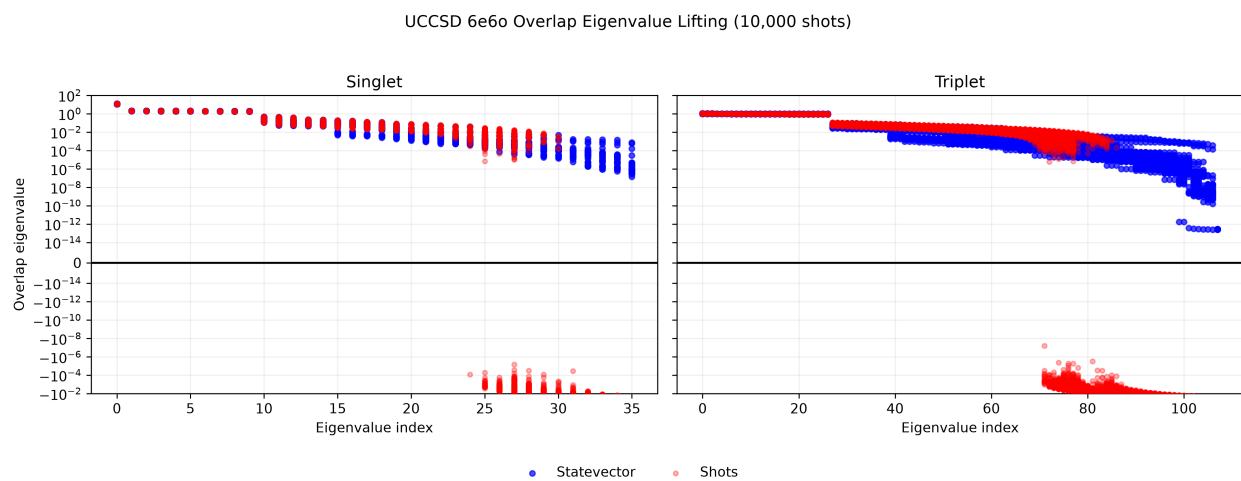


Figure S5: Finite-shot lifting of UCCSD overlap eigenvalues in the 6e6o active space.

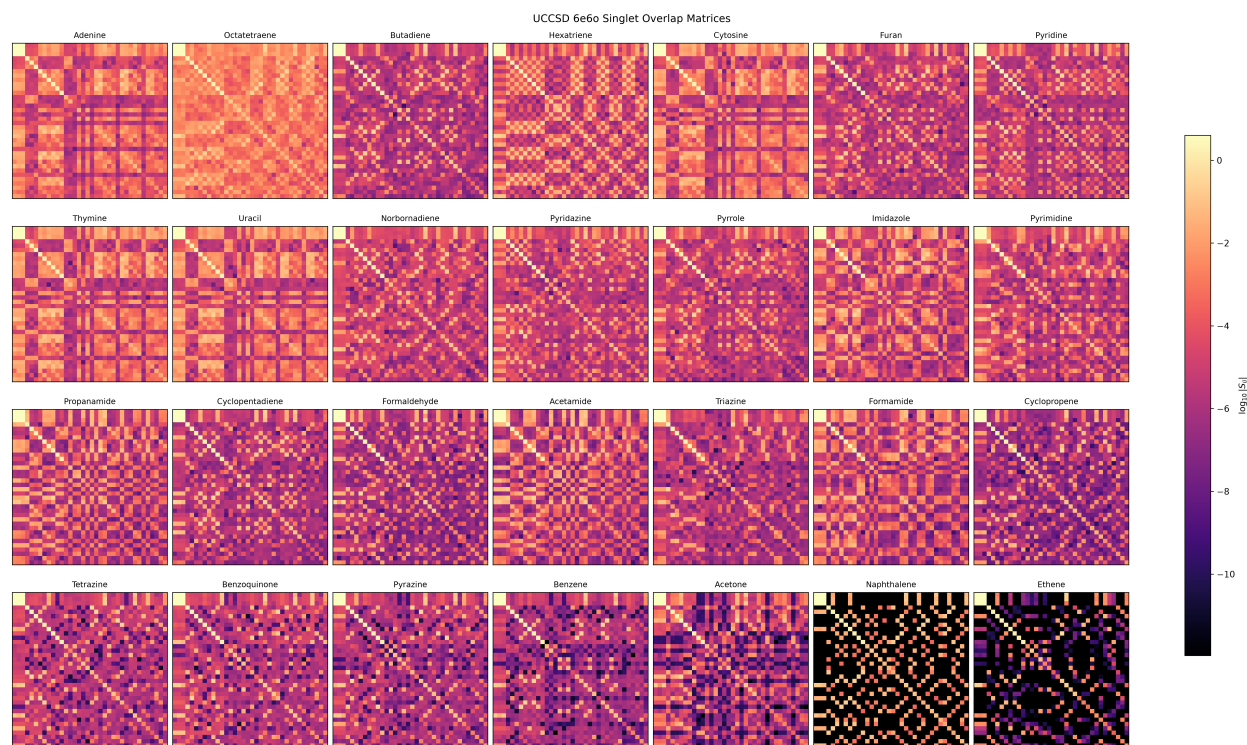


Figure S6: Full 28-molecule singlet overlap heatmaps comparing statevector and shot-based results.

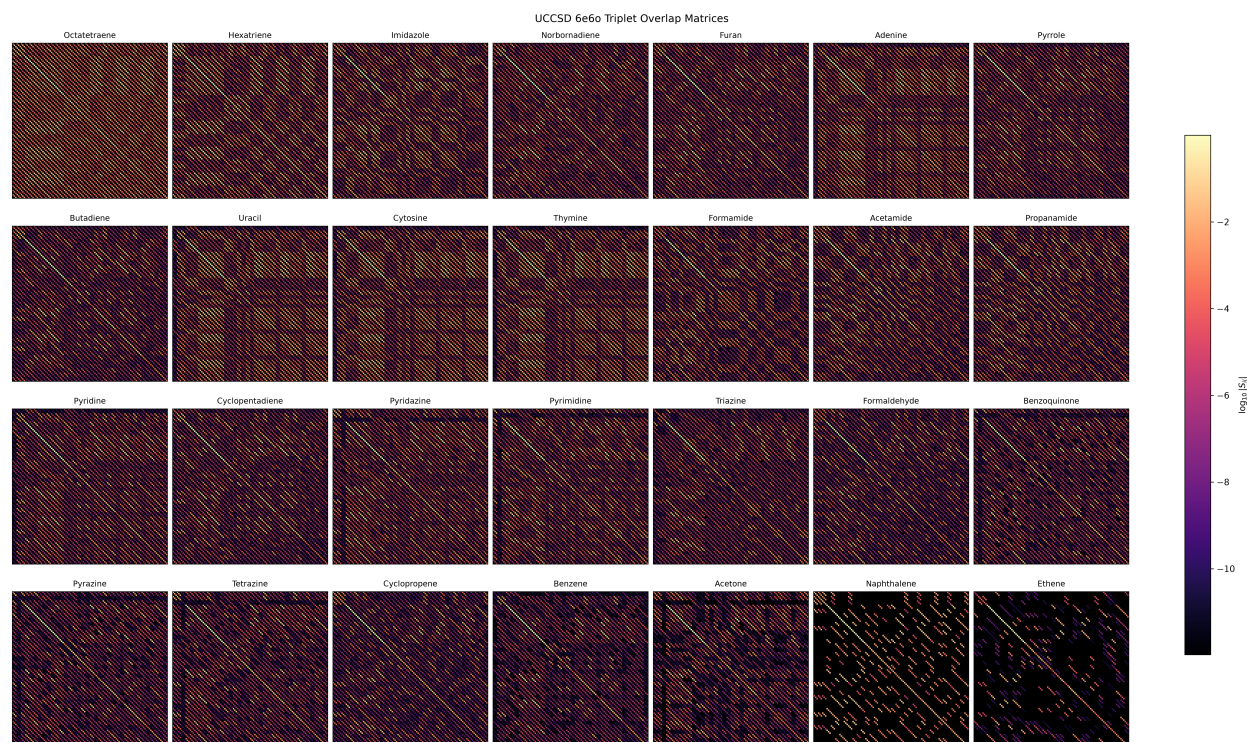


Figure S7: Full 28-molecule triplet overlap heatmaps comparing statevector and shot-based results.

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